
Investigating Impact of Quantum Ansatz Complexity in Quantum Hybrid Fusion Models to Estimate Protein-Ligand Binding Energy

Anonymous Author(s)

Affiliation

Address

email

Abstract

Calculating binding energy for proteins is an essential step in evaluating efficacy of potential drugs. It is computationally expensive and difficult, especially for large molecules, due to the number of complex interactions that must be considered between proteins and ligands. One promising direction is quantum machine learning to augment classical models. This paper presents a fusion model that extracts features from a 3D-CNN and an SGCNN, then feeds them into a quantum layer simulated using Qiskit or PennyLane to predict binding energy. The main contributions of our work are quantifying the impact of replacing the unitary circuit and analyzing variability across circuit choices, comparing fixed and parameterizable circuits, and quantifying effects of gate composition. The findings were, that based off of the code repositories that were used for this work, there is little difference and variability between fixed and parameterizable circuits at this scale, and furthermore, no statistically significant change in predictability of protein-ligand binding affinity through swapping 1-5 layers with unitary circuit layers; it remains possible that they are learning to fit around the noise. As such, it remains even more important to ensure that QNNs are not just learning to fit around the noise, and we have robust benchmarks for detecting when this happens.

1 Introduction

Calculating binding energy for proteins is an essential step in evaluating efficacy of potential drugs (7). Accurate prediction of protein-ligand binding free energy remains both a major challenge and a major promise in computational drug design, since binding affinity estimation is needed to evaluate and optimize drug candidates before synthesis and experimental testing begins (7; 9; 4). However, this process is computationally expensive and difficult, especially for large molecules, due to the number of complex interactions that must be considered between proteins and ligands. There has been a substantial amount of previous research using purely classical models to calculate ligand-protein binding energies (1; 5; 9; 7), as well as more recent work using quantum-classical hybrid models (3; 2; 13; 12).

Domingo et al. used a hybrid 3D-CNN model and demonstrated the potential for quantum machine learning on protein-ligand binding affinity prediction by replacing the penultimate layer with a quantum circuit (3). A later study in 2024 (2), aimed at simplification of QNN layer by building upon 3DCNN & SGCNN fusion model from Jones et al.(7) and replaced the penultimate fusion layer with different ansatzes (2).

One promising direction is quantum machine learning to augment classical models. This paper presents a fusion model that extracts features from a 3D-CNN and an SGCNN, then feeds

them into a quantum layer simulated using Qiskit or PennyLane to predict binding energy. The main contributions of our work are quantifying the impact of replacing the unitary circuit and analyzing variability across circuit choices, comparing fixed and parameterizable circuits, and quantifying effects of gate composition. We find that in some cases, the surrounding classical network may learn to become insensitive to the quantum layer. In other words, the unitary circuit acts primarily as a perturbation mechanism rather than improving the accuracy or performance of the energy prediction. If this is the case, changing the circuit architecture, increasing its complexity, or replacing additional classical layers with quantum circuits should have limited impact on predictive performance.

Our main findings were that within our tested scope, replacing layers with unitary circuits had minimal impact on the overall NNs, which just fit around the random unitary circuit layers.

2 Methodology

2.1 Brief overview

Protein-ligand complexes from the PDB were preprocessed using standard techniques including hydrogenation and charge assignment. We applied one-hot encoding to atomic hybridization, resulting in 3 extra features. The data was voxelized into a $22 \times 48 \times 48$ grid, where there are 22 channels for features, and the molecules have been centered in a 3D grid with resolution of 48 angstroms in the x, y, and z dimensions. Further details about the 3D-CNN and SG-CNN models we used and how the voxelized protein-ligand data was processed can be found in our appendix.

2.2 Quantum layer integration

The quantum network used in our exploration was constructed by using a fully connected quantum layer from the Ingenii repository (6). We tested random quantum reservoir circuits from the G3 family, and compared comparative performance. We generated 100 random gates, and then preselected the top 25 using an expressibility criterion (RFD), then compared their predictive performance, using R^2 and adjusted R^2 values. RFD was inspired by prior work on evaluating expressibility and entangling capability of parameterized quantum circuits (11). As the Domingo paper (3) had found diminishing returns at approximately 300 gates, our circuit size was also set to 300 for some experiments, and then lower for others. Our experiments used 6 qubits as the primary configuration, with gate counts of 30, 100, 300, and 1000 tested in a gate variance study.

3 Results and discussion

Comparing unitary circuits performance, along with quantum kernels

Overall, replacing one layer in the 3D/SG CNN fusion model with a random unitary circuit does not significantly change the R^2 of the model in predicting the protein-ligand binding affinity. As an extension, the authors replaced multiple layers in the 3D/SGCNN fusion network. One through five layers were replaced with an independent fixed random unitary circuits, each containing three G3 gates. As shown in Figure 1(a), the resulting holdout R^2 values remained closely clustered and no statistically significance depend on the number of replaced layers was observed. Thus, these results show that the tested random unitary substitutions preserve the predictive behavior of the fusion model, while the simple circuit descriptors, such as depth, gate counts, and 1st-order gate transitions do not explain the performance variability among individual circuits.

Although the differences in the violin plot look large, the performance of the circuits is largely unchanged across the different fixed and parameterizable circuits (with 3, 10, and 30 gates), as seen in Figure 1(b). Although the violin plots show some visible variation, the substantial overlap between conditions indicate that neither additional gates nor parametrization yielded a reliable improvement over compact fixed reservoirs. As such, this finding suggests that even with just a small random unitary circuit gates in the replaced layer, the gate composition does not significantly impact the performance of the selected 3D/SGCNN neural-network layer without substantially changing the binding-affinity prediction performance.

To determine whether the choice of a random unitary circuit produced various, measurable

86 difference in predictive performance, we made scripts to analyze the G3-family random reservoir circuits patterns between the top and bottom quartile-performing unitary circuits, ranked by adjusted R^2 .
87 We also aggregated gate-composition statistics and local gate-order statistics and for each circuits, we
88 computed features such as the number of H, T, and CX gates, circuit depth, gate ratios, frequency of
89 gate order (e.g. CX after H or vice versa), and first-order transition frequencies between adjacent gates.
90
91 As shown in Figure 1(c–d), we found across the tested descriptors, no gate-count, gate-
92 ratio, or depth feature that there is a statistically significant separation between high and low
93 performing circuits. Similarly, the first-order transition analyses showed that only the small shifts in
94 adjacent-gate patterns with no transition type that provides a statistically significant information for
95 the performance differences. These results suggests that the circuit performance is not well explained
96 by simple composition-level features or by local two-gate ordering patterns.
97

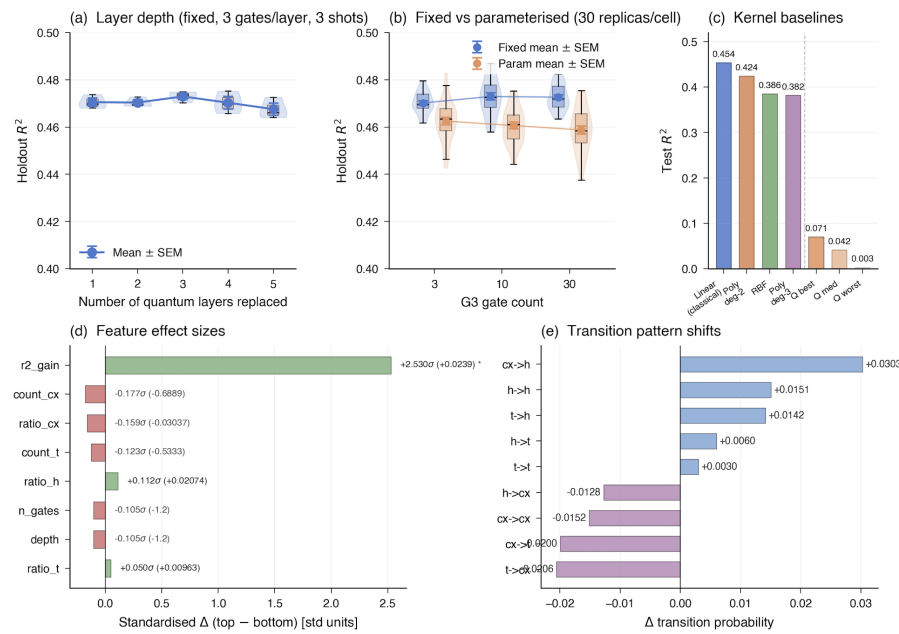


Figure 1: Overview of experiments testing sensitivity to the quantum layer. (a) Holdout R^2 as one through five layers are replaced with independent fixed random unitary circuits containing three G3 gates per layer. Performance remains closely clustered across the number of replaced layers. (b) Comparison of fixed and parameterizable circuits with 3, 10, and 30 G3 gates. Neither additional gates nor parametrization yielded a reliable improvement. (c) Gate-composition features comparing higher- and lower-performing circuits. (d) First-order transition-frequency differences between higher- and lower-performing circuits. Together, the experiments show limited sensitivity to the number, parametrization, and simple composition of the tested quantum circuits.

98 Thus, as shown in Figure 2, the quantum linear kernel under performs every classical baseline by a
99 wide margin, even when using the best-performing circuit identified by RFD ($R^2 = 0.0708$ vs. $R^2 =$
100 0.3855 – 0.4536 for classical kernels). Note that the gap between the best and worst circuits (0.0708
101 vs. 0.0032) is small relative to the gap between any quantum circuit and the classical baselines,
102 indicating that the circuit selection has a limited influence on kernel quality in this workflow. This
103 result reinforces our central finding from the ablation study.
104

105 The apparent performance gain is then observed when the quantum reservoir is embedded
106 in a skip-connected architecture is attributable to the classical pathway, not the quantum feature map
107 itself. So when evaluated by itself without a classical skip connection, the quantum kernel captures
108 only a small fraction of the variance in binding affinity, suggesting that the 6-dimensional quantum
109 encoding retains too little of the original 153-dimensional signal to serve as an effective standalone
110 representation.

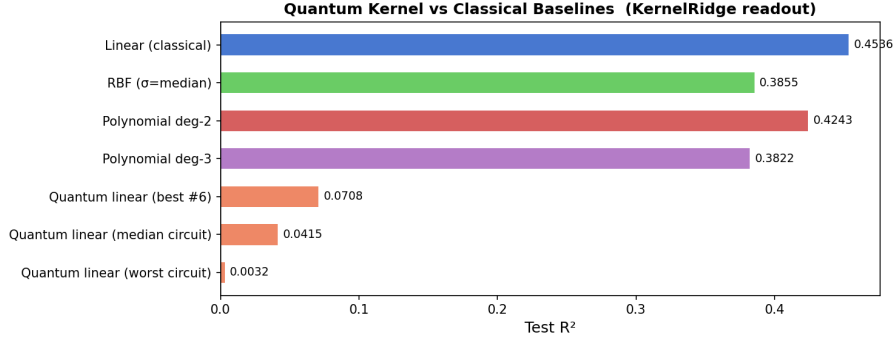


Figure 2: Comparison of classical kernel baselines against quantum linear kernels derived from the best, median, and worst circuits in the sweep (readout via KernelRidge). All classical kernels (linear, RBF, polynomial) achieve R^2 in the 0.38–0.45 range, while every quantum kernel variant remains below $R^2 = 0.08$, regardless of circuit quality.

4 Limitations and Next Steps

Overall, our contribution is to combine a 3D-CNN and SG-CNN fusion architecture with a fixed quantum reservoir layer and to study the variability introduced by different random G3-family circuit choices. This allows us to test whether circuit composition itself has a measurable effect on binding-affinity prediction performance, while keeping the quantum layer non-trainable and computationally manageable; making the layer trainable could be a potential next step.

The experiments suggest that there was not a substantial difference in R^2 and predictability patterns between the runs without the random unitary circuits, and the runs with the random unitary circuits that had 3, 10, 30, 100 gates, as well as those respective gates that were also parameterizable. This heavily suggests that only a few gates are required to replace one layer in a neural network without much difference. In addition, replacing one layer, and multiple layers (up to 5) were statistically insignificant from replacing just one layer. One possible explanation for this insensitivity is that the classical components of the hybrid model learn to compensate for the transformation introduced by the quantum layer. Following, the random unitary circuit contributes a manageable perturbation that the surrounding network can learn around. The random unitary circuit does not contribute to a structure that helps improve predictive performance. Although a next step would be to run more specific experiments, the observed invariance across circuit parametrization, gate count, and number of replaced layers from our experiments is consistent with this hypothesis.

Another limitation stems from the dimensionality reduction performed in our quantum encoding step. Since we compressed a 153-dimensional classical feature vector down to 6 dimensions, the observed insensitivity to the quantum layer cannot be definitively attributed to the classical network compensating for and absorbing the quantum transformation, as we hypothesized. Alternatively, or additionally, it could reflect a loss of information incurred by the compression itself, independent of any specific property of the quantum circuits we tested. Separating these two possible explanations matters because the hypothesis: that the classical architecture rather than the quantum layer could be responsible for the observed performance, is only defensible once dimensionality reduction has been ruled out as a confounding variable. To properly characterize this, we propose evaluating matched classical controls performing an equivalent 153-to-6 dimensional reduction, such as a random linear projection, a PCA-based projection, or a trainable dense bottleneck layer of the same width, and compare their standalone and fused predictive performance against the quantum encoding.

Coming back together, the ablation and kernel-based analyses converge on the same conclusion in which the quantum component’s contribution is largely absorbed by the surrounding classical architecture than the quantum feature map itself. Future work could build on using the random unitary circuit replacement strategy and directly calculating the exact hardware requirements required for it to provide a compute-efficiency advantage. Another interesting question is how the tradeoff changes based on the volume of data, as well as the dimensionality and scarcity of it.

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Appendix

A Binding affinity models: notable mentions

There has also been a subset of approaches in the classical and hybrid models that are worth mentioning.

Of these, we tried running DeepChem’s Atomic Convolutional Network, and replicated many others, but found that the Domingo and Jones methodologies were well documented, and more modular, thus easier to run a greater number of experiments on.

A.1 Classical Models

A recent paper, published in April 2025, used a classical Super Learner model that combines 10 different machine learning algorithms, including random forest. A separate paper used linear regression to determine a free energy function. Another model with comparable performance is the Atomic Convolutional Neural Network (ACNN) (5), which was implemented in the open-source molecular machine learning package DeepChem. Also, a quantum mechanics/molecular mechanics (QM/MM) paper achieved a Pearson correlation of 0.74 on six different proteins and 109 ligands (9).

A.2 Hybrid Quantum Models

Kirsopp et al. (8) used Density Matrix Embedding Theory (DMET) and the Variational Quantum Eigensolver (VQE) to quantify protein-ligand interactions for a series of β -secretase inhibitors. A 2025 bioRxiv preprint by Roosan et al. (10) reported quantum-enhanced modeling approaches for protein-ligand binding prediction, including a hybrid quantum-classical model evaluated on PDBbind-derived data. Yang (13) explored quantum machine learning for predicting binding free energies in structure-based virtual screening using parameterized quantum gates. In a separate preprint, Yang (12) proposed a quantum convolutional model for structure-based virtual screening using PDBbind v2020.

Study	Model	R	R^2
Audie & Scarlata, 2007 (1)	Empirical free energy function	0.73	0.533
Gomes et al., 2017 (5)	ACNN, temporal split	0.727	0.529
Gomes et al., 2017 (5)	GRID-RF, temporal split	0.697	0.486
Molani & Cho, 2024 (9)	QM/MM	0.74	0.55
Jones et al., 2021 (7)	3D-CNN	0.666	0.443
Jones et al., 2021 (7)	SG-CNN	0.723	0.523
Jones et al., 2021 (7)	Fusion	0.810	0.656
Domingo et al., 2023 (3)	Hybrid quantum 3D-CNN	~ 0.699	0.489
Domingo et al., 2024 (2)	Hybrid quantum fusion model	–	–
Yang, 2025 (13)	Parameterized quantum circuit	0.650	0.423
Yang, 2025 (12)	Quantum convolutional neural network	0.694	0.482

Table 1: Summary of selected prior models for binding affinity prediction. Reported values come from studies using different datasets, splits, and evaluation settings, so they should be interpreted as approximate points of comparison.

B Hyperparameter Experiments

For 3D-CNN:

- Best $R^2 = 0.4589$ at learning rate 0.004
- Best decay rate = 0.95

For SG-CNN:

552 • Best $R^2 = 0.4007$ at learning rate 0.004

553 C Initial Explorations

554 Before we begun implementing the ML pipelines, we explored the shape of the data ourselves, and
555 have included it here for those interested.

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557 Figure 5 shows the relative atom locations for the protein, ligand, and ligand pocket. Compared with
558 the full protein structure, the ligand and pocket regions are more sparsely populated, which reduces
559 irrelevant spatial information in the model input.

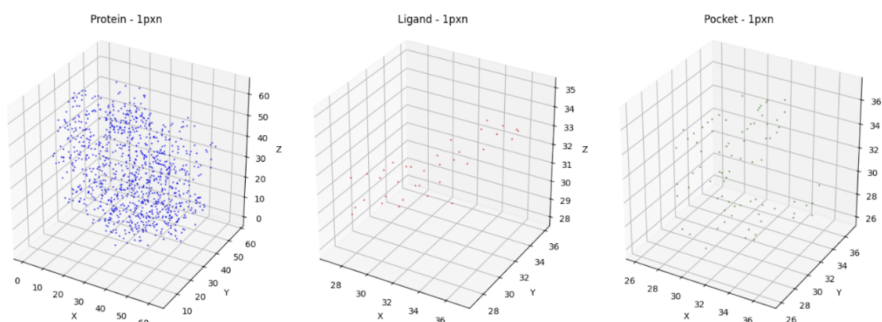


Figure 3: Visualization of atom locations for the protein, ligand, and ligand pocket. The ligand and pocket are more sparsely populated than the full protein structure, which suggests that these regions contain less spatial noise for downstream feature extraction.

560 As shown in Figure 6, the binding energy distribution was approximately centered and did not exhibit severe skew, suggesting that the target values were reasonably well scaled before model training.

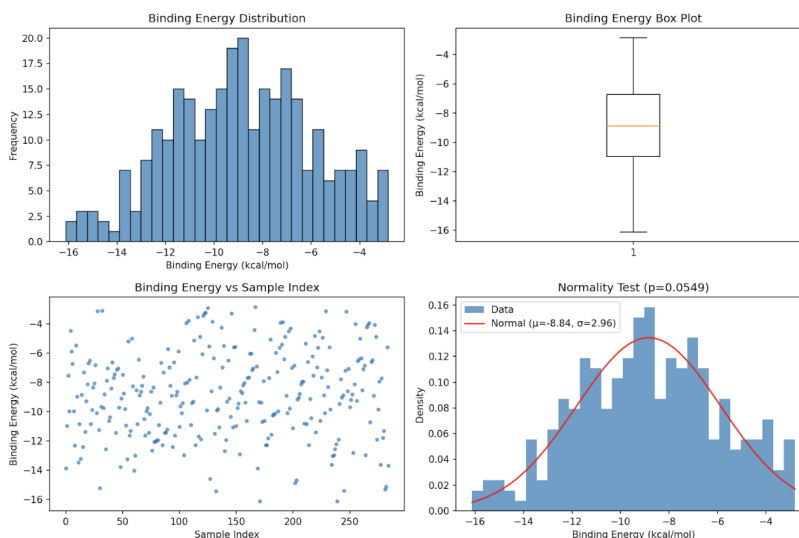


Figure 4: Distribution of binding energy values in the dataset. The distribution is approximately centered and does not show extreme skew, suggesting that the target values were already relatively well normalized before model training.

561

D Data Processing

- **Element type(9):** one-hot encoding of B, C, N, O, P, S, Se, halogen, or metal.
- **Structural properties(5):** 5-bit vector encoding(1 present, 0 absent) representing (hydrophobic, aromatic, acceptor, donor, ring).
- **Atom hybridization(3):** one-hot encoding of (1, 2, or 3)
- **Atom characteristics(2):** Heavy valence atom bonds and number of bonds with other heteroatoms.
- **Molecule type(1):** to indicate protein atom(-1) versus ligand atom(+1).
- **Partial charge(1):** Representing the continuous atom level charge to encode local electrostatic variation for protein-ligand interactions
- **Van der Waals radius(1):** Representing the continuous atom level of its effective nonbonded atomic radius, helping the model capture its molecular shape, packing constraints, etc within the binding pocket

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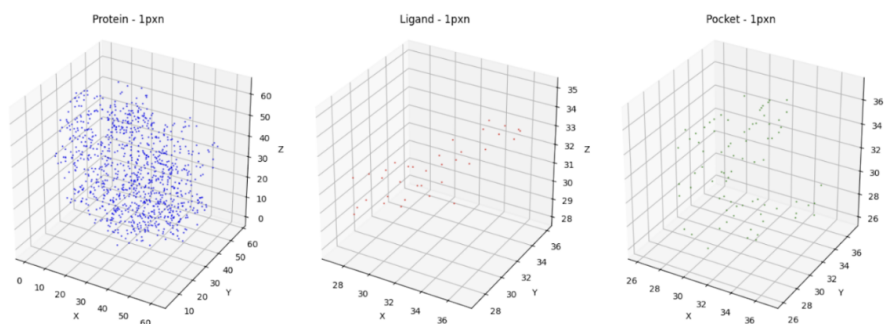


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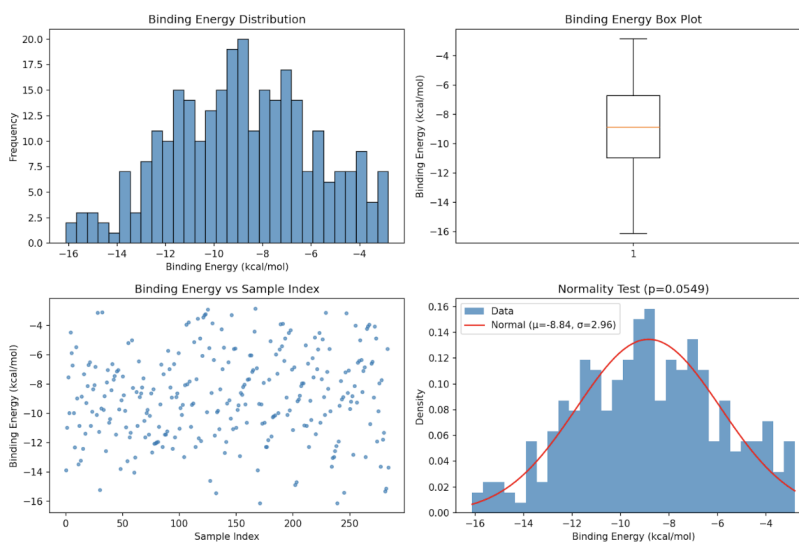


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